

Literature Status for the Discrete Lipschitz-Free Approximation Property Question in arXiv:2005.06555

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Status

This note records that the discrete-metric-space approximation-property question from Albiac–Ansorena–Cuth–Doucha, arXiv:2005.06555, is answered positively by Albiac–Ansorena–Bima–Cuth, arXiv:2506.09786.

No new theorem is claimed here. The purpose of this packet is to clear an open-problem signal from the run queue and identify the exact later theorem which settles it.

Source Question

In the Open Problems section of arXiv:2005.06555, the authors ask:

Let $\mathcal{M}$ be a discrete metric space. Does $\mathcal{F}(\mathcal{M})$ have AP? Or, more generally, does $\mathcal{F}_p(\mathcal{M})$ have AP for every $p \in (0, 1]$?

The later paper cites this as AACD21, Question 6.3.

Later Answer

The abstract and introduction of arXiv:2506.09786 state that the paper proves the approximation property for Lipschitz free p -spaces over discrete metric spaces and thereby answers the question raised in arXiv:2005.06555.

The theorem-level statement is arXiv:2506.09786, Theorem `thm:discreteCase`: if $\mathcal{M}$ is a complete metric space with at most finitely many accumulation points, then $\mathcal{F}_p(\mathcal{M})$ has the approximation property for every $p \in (0, 1]$. The line immediately before the theorem says that this answers AACD21, Question 6.3, in the positive.

In the proof of the theorem, the authors invoke Proposition 4.1 of arXiv:2005.06555, and the comment before Question 6.3 there, to reduce to the case where $\mathcal{M}$ is a complete discrete metric space. They then use their compact-reduction principle and finite-rank approximants to prove the approximation property. Thus the source question is not merely implied by an agent-side reformulation: the supporting paper explicitly labels and answers the same question.

Scope

The resolved scope is exactly the discrete metric space AP question for $0 < p \leq 1$. This packet does not settle the neighboring countable proper metric approximation property question, the problem

asking for a metric space $\mathcal{M}$ with $\mathcal{F}_p(\mathcal{M})$ failing AP, or the other isomorphism/basis/net questions listed in the same Open Problems section of arXiv:2005.06555.

References

1. F. Albiac, J. L. Ansorena, M. Cuth, and M. Doucha, *Lipschitz free spaces isomorphic to their infinite sums and geometric applications*, arXiv:2005.06555; Trans. Amer. Math. Soc. 374 (2021), 7281–7312.
2. F. Albiac, J. L. Ansorena, J. Bima, and M. Cuth, *Lipschitz free p -spaces for $0 < p < 1$ in the light of the Schur p -property and the compact reduction*, arXiv:2506.09786.